Hybrid Post-Quantum Key Exchange for WireGuard

Integrating ML-KEM into a Production Rust Userspace Implementation

Mihai-Cristian Farcaș

Babeș–Bolyai University
Faculty of Mathematics and Computer Science

Bachelor's Thesis Defense | July 2026

Supervisor: Prof. Dr. Darabant Sergiu Adrian

Roadmap

1. The problem – why WireGuard needs post-quantum key exchange
2. Two approaches – PSK-slot baseline & full hybrid
3. The hybrid handshake, security & implementation
4. Evaluation & conclusions – then a live demo

The problem: a quantum clock, and a protocol that can't be changed

- WireGuard's key exchange rests entirely on X25519 – which Shor's algorithm breaks once a large quantum computer exists.
- **Harvest-now, decrypt-later:** adversaries record encrypted traffic *today*, decrypt it *later*. For long-lived secrets, migration is urgent *now*.
- NIST finalised **FIPS 203 (ML-KEM)** in 2024; classical key exchange **deprecated by 2030**, disallowed by 2035.

But WireGuard is deliberately rigid

One fixed cipher suite, **no negotiation** – a security feature (kills downgrade attacks), but no slot to drop a new algorithm into. Two natural handles: the **PSK slot**, and the **Noise IK handshake** itself.

Research question and two approaches

The question

Can WireGuard's handshake be extended with **hybrid post-quantum key exchange (X25519 + ML-KEM-768)** while preserving its security properties and staying viable on **resource-constrained hardware**?

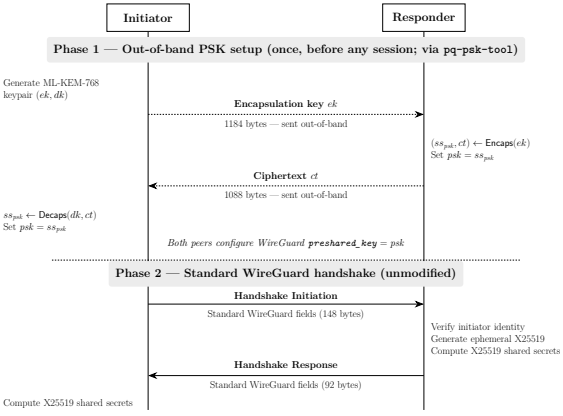
A. PSK-slot baseline

- One-time ML-KEM exchange out-of-band → fill the PSK slot.
- **Zero** WireGuard changes; works with any client.
- The *lower bound* on engineering effort – a controlled comparison point.

B. Full hybrid **(the contribution)**

- ML-KEM-768 *inside* the Noise IK handshake, every session.
- Fresh KEM keypair per handshake.
- Target: **BoringTun** (Cloudflare's production Rust userspace WireGuard).

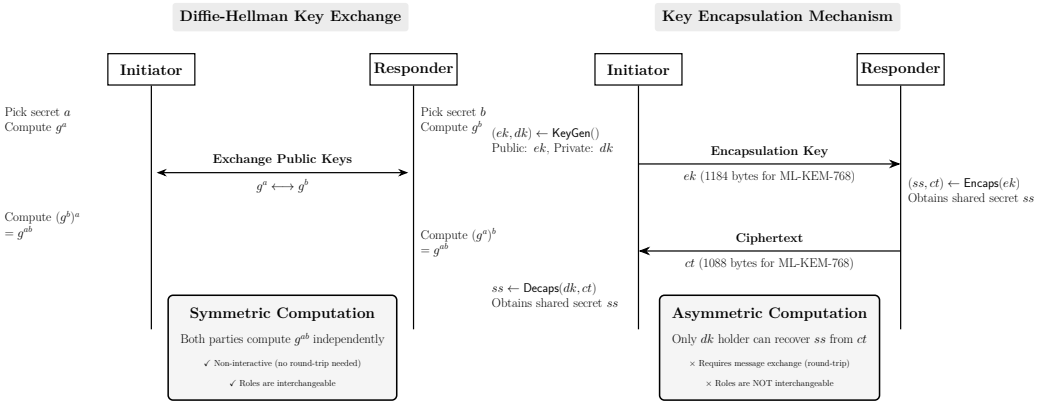
Backup – the PSK-slot baseline



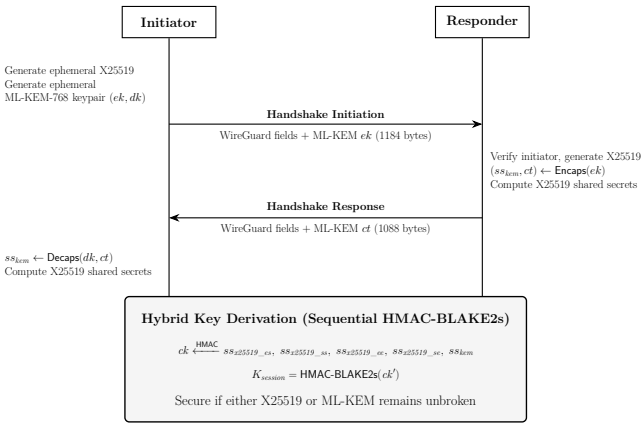
Why ML-KEM-768, why hybrid

- **ML-KEM-768:** The general-purpose default recommended by NIST/IETF; balances size vs security.
- **Hybrid, not pure-PQ:** the MLWE assumption is young; pairing with X25519 means a future lattice break still leaves sessions protected. NIST and IETF recommend hybrid for the transition.
- **vs related work:** PQ-WireGuard is pure-PQ on prototypes with non-standard KEMs; Rosenpass/ExpressVPN carry PQ secrets through the PSK slot. This work modifies the Noise handshake directly, on production-ready code, with standardised ML-KEM – complementary to the formal Hybrid-WireGuard proof (Lafourcade et al.).

A KEM is not a Diffie–Hellman



The hybrid handshake: ML-KEM rides the existing two messages



- **Msg 1:** WG fields + ML-KEM *ek* (1,184 B)
- **Msg 2:** responder Encaps + *ct* (1,088 B)
- Initiator Decaps → shared ss_{kem}; **no extra round trips**

Combining secrets: after the four X25519 DHs, one extra HMAC-BLAKE2s step folds in ss_{kem} (WireGuard's *own* KDF); *ek*/*ct* bind into the transcript hash.

ek/*ct* sit *before* the MACs ⇒ the anti-DoS cookie covers the PQ payload. Transport untouched – throughput stays identical.

Security analysis

Hybrid-secure

The session key is indistinguishable from random if **X25519 *or* ML-KEM** is secure (sequential chaining / concatenate-then-KDF; HMAC-BLAKE2s as a PRF).

- **Post-quantum forward secrecy:** fresh ML-KEM keypair *per handshake* ⇒ past sessions stay safe even if X25519 later falls.
  The PSK-slot baseline cannot match this directly: a static PSK, one compromise exposes many sessions.
- **Authentication stays classical** (X25519 static keys) – *deliberate*: it cannot be broken *retroactively*, and NIST prioritises key-exchange migration over signatures. ML-DSA is future work.

Implementation

- All PQ code behind a Cargo `pq` feature flag ⇒ **a vanilla build is byte-for-byte identical to upstream**.
- `ml-kem` crate (RustCrypto): pure-Rust, `no_std`, NIST KAT-tested ⇒ **no C FFI**, runs on the Pi.
- New message types **5**/**6**; parser dispatches on type+size; a vanilla build cleanly rejects them (`InvalidPacket`).

Extended BoringTun's own Criterion benches & `noise::tests` – full PQ handshake, end-to-end transport, vanilla-rejects-PQ.

... and we'll see it running in the live demo.

Evaluation – per-primitive & handshake cost (Criterion, 95% CI)

Per-primitive (µs)

Operation	M4	Pi 5
ML-KEM-768 KeyGen	24.13	81.66
ML-KEM-768 Encaps	20.82	82.77
ML-KEM-768 Decaps	28.03	115.15
X25519 KeyGen	8.77	58.00
X25519 DH	20.39	192.81

Full handshake (µs)

Config	M4	Pi 5
Vanilla	150.8	1312
PSK-slot	151.0	1312
Hybrid	**247.3**	**1643**
overhead	**+64%**	**+25%**

- **Surprise:** on the Pi one X25519 DH (193 μs) is *slower* than any ML-KEM op – `x25519-dalek` lacks NEON; ML-KEM's NTT maps to the integer pipeline.
- **Size is the real cost:** handshake **240 B** $\rightarrow$ **2,512 B**, yet both messages fit a 1,500 B MTU in one datagram (confirmed over Wi-Fi).
- **Footprint negligible:** binary +65 KiB, RSS +0.16 MiB.
- **Throughput identical** – same data plane; you'll see it live.

Conclusions

Contributions

1. First FIPS 203-compliant *hybrid* WireGuard on a production Rust codebase.
2. A PSK-slot baseline as a controlled comparison point.
3. A hybrid Noise IK handshake with *post-quantum forward secrecy*.
4. Evaluation across two ARM tiers (Apple M4, Raspberry Pi 5).

Bottom line: hybrid PQ WireGuard is **practical today** – NIST-aligned, userspace-deployable, sub-millisecond overhead, data plane untouched.

Future work: MTU strategies (segmentation / PMTUD); ML-DSA authentication; a formal proof; ML-KEM parameter agility.

Thank you.

Questions?

Mihai-Cristian Farcaș | Supervisor: Prof. Dr. Darabant Sergiu Adrian

Backup – MTU & interoperability

- Hybrid initiation 1,332 B; fragmentation only on paths below ∼1,360 B (e.g. IPv6's 1,280 B minimum).
- **Lowering the WireGuard MTU does *not* help** – that shrinks data packets, not the fixed-size handshake. The fix must live in the handshake (app-level segmentation / PMTUD – future work).
- Interop: a vanilla peer drops PQ types 5/6 at dispatch (`InvalidPacket`); no crash, but no auto-fallback – both peers must be hybrid-capable. Capability negotiation is engineering, not fundamental.

Backup – combining the secrets (key derivation)

One extra mixing step, using WireGuard's own KDF

After the four X25519 secrets are mixed in as usual, fold in ss_{kem}:

$$temp = \text{HMAC-BLAKE2s}(ck,\ ss_{kem}), \qquad ck' = \text{HMAC-BLAKE2s}(temp,\ \texttt{0x01})$$

ek and *ct* are also bound into the transcript hash *h*.

- **Reuses WireGuard's HMAC-BLAKE2s** – no new HKDF/SHA-256 introduced; keeps the security argument clean.
- Transport phase is byte-for-byte unchanged – ChaCha20-Poly1305 untouched. *This is why throughput is identical.*

Backup – throughput & footprint detail

- **Throughput unchanged:** ∼470 Mbps loopback, ∼80–85 Mbps cross-device; vanilla vs hybrid CIs overlap (same ChaCha20-Poly1305 data plane). Treat the table as a correctness check.
- **Footprint negligible:** binary **+65 KiB** (+5.7%), peak RSS **+0.16 MiB** – a non-issue even on the Pi.
- **Packet size:** handshake grows **240 B** → **2,512 B** (+2,272 B); both messages fit a 1,500 B Ethernet MTU in a single datagram, confirmed over real Wi-Fi, no fragmentation.